Logan Frank

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Education

Ohio State University — Columbus, OH
Ph.D. in Computer Science and Engineering — August 2019 - Present
Advisor: Prof. Jim Davis — Expected Spring 2025

Wright State University — Dayton, OH
B.S. in Computer Engineering — August 2015 - May 2019

Experience

Ohio State University, Computer Vision Laboratory — Columbus, OH
Graduate Research Associate — August 2019 - Present

Data-Free Knowledge Distillation:

- Developed a novel and completely privacy-preserving data-free knowledge distillation approach that is unique in the sense that it does not require a generator network or any other means of synthesizing examples from the teacher model
- Leveraged procedural image programs to generate synthetic data which was combined with heavy data augmentation and adversarial attacks to successfully transfer knowledge from a trained teacher model to a freshly initialized student
- Demonstrated state-of-the-art performance on student models for multiple image classification datasets

Improvements to Batch Normalization (ECCV 2022):

- Observed that the learned affine transformation parameters of batch normalization do not alter much from their initial state and can allow overly large values (output from the preceding normalization) to be passed forward
- Proposed an alternative initialization and updating strategy to the scale parameter of batch normalization, which resulted in statistically significant accuracy gains over the default initialization and other related approaches

Agricultural Crop Stress Recognition (WACV 2021):

- Collaborated with the Department of Food, Agricultural, and Biological Engineering at Ohio State to apply a confidence grounded hierarchical inference approach to plant stress identification

Air Force Research Laboratory — Dayton, OH
Graduate Research Intern (Mentor: Dr. Christopher Liberatore) — Summer 2023

- Created a feature augmentation method for addressing long-tail imbalanced classification (work is under U.S. CUI)
- Investigated existing long-tail imbalanced classification approaches applied to DoD-specific data

Air Force Research Laboratory — Dayton, OH
Graduate Research Intern (Mentor: Mr. Christopher Menart) — Summer 2019

- Researched into utilizing knowledge graphs for creating more explainable, generalizable, and robust neural networks

Publications

L. Frank, J. Davis
"Data-Free Knowledge Distillation Using Adversarially Perturbed OpenGL Shader Images"
arXiv:2310.13782 [cs.CV] (October 2023)

J. Davis, **L. Frank**
"Revisiting Batch Norm Initialization"
European Conference on Computer Vision (2022)

L. Frank, C. Wiegman, J. Davis, S. Shearer
"Confidence-Driven Hierarchical Classification of Cultivated Plant Stresses"
IEEE/CVF Winter Conference on Applications of Computer Vision (2021)

Z. Daniels, **L. Frank**, C. Menart, M. Raymer, P. Hitzler
"A Framework for Explainable Deep Neural Models Using External Knowledge Graphs"
SPIE Defense and Commercial Sensing: AI and ML for Multi-Domain Operations Applications Track (2020)

Technical Skills

Languages & Tools: Python, Bash, Git, Singularity containers, PBS job scheduler, Slurm job scheduler, LaTeX

Professional Service, Awards, and Activities

Reviewer: CVPR'24, ICCV'23, CVPR'23, ECCV'22, CVPR'22
SMART Scholarship Recipient, United States Department of Defense — 2019
NCAA Division 1 Athlete - Swimming, Wright State University Athletics — August 2015 - May 2017